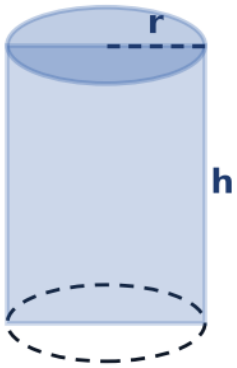
Distributions and Maximum Likelihood

ClipSAT Printable Practice Suite (clipsat.org) • Topic 5.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Distributions & Maximum Likelihood Practice Exercises

Use the method of Lagrange multipliers to find the maximum volume of a rectangular box with faces parallel to the coordinate planes inscribed in the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$.



2 [Foundational] • Target: Distributions & Maximum Likelihood Practice Exercises

A temperature distribution is given by $T(x, y, z) = \frac{120}{1 + x^2 + 2y^2 + 3z^2}$. At the point $P(1, 1, 1)$, find: (a) the direction of steepest temperature increase, (b) the maximum rate of increase, and (c) the equation of the tangent plane to the isothermal level surface passing through P.

3 [Foundational] • Target: Distributions & Maximum Likelihood Practice Exercises

Level 3 · AdvancedWater flows into an initially empty tank at a rate $R(t) = 120 - 4t$ liters/hour and leaks out at a rate $L(t) = 3t$ liters/hour for $0 \le t \le 24$ hours. Find the maximum volume of water in the tank and the time at which the tank becomes completely empty.

4 [Foundational] • Target: Distributions & Maximum Likelihood Practice Exercises

[Surface Integrals, the Divergence Theorem & Stokes' Theorem · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

5 [Foundational] • Target: Distributions & Maximum Likelihood Practice Exercises

[Triple Integrals & Cylindrical/Spherical Coordinates · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

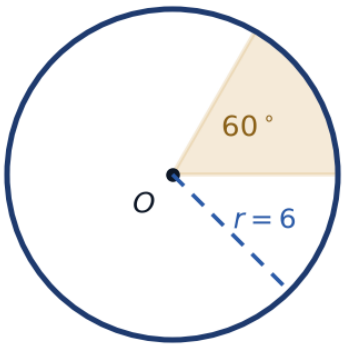
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6 [Foundational] • Target: Distributions & Maximum Likelihood Practice Exercises

[Vector Fields & Line Integrals · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

7 [Foundational] • Target: Distributions & Maximum Likelihood Practice Exercises

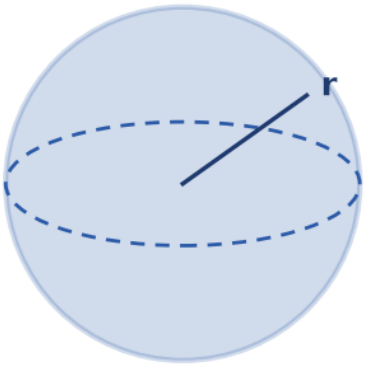
Let $z = f(x, y)$ be a differentiable function, with polar coordinates transformation $x = r\cos\theta$ and $y = r\sin\theta$. Express $\frac{\partial z}{\partial r}$ and $\frac{\partial z}{\partial \theta}$ in terms of $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$, and prove that: $\left(\frac{\partial z}{\partial r}\right)^2 + \frac{1}{r^2}\left(\frac{\partial z}{\partial \theta}\right)^2 = \left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 = ||ablaz||^2$.



Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Distributions & Maximum Likelihood Practice Exercises

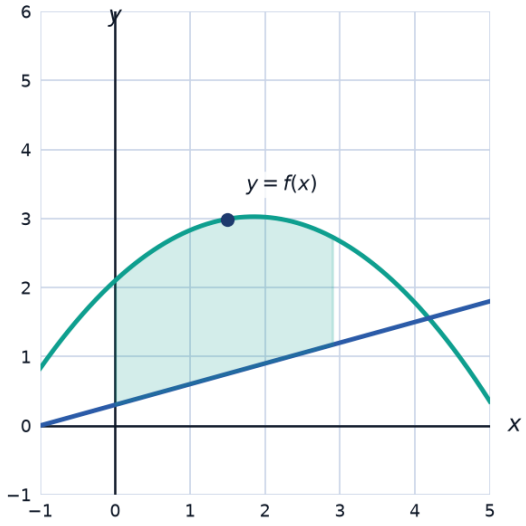
Find the total mass and the z-coordinate of the center of mass of the upper solid hemisphere $x^2 + y^2 + z^2 \le R^2$, $z \ge 0$, whose density is proportional to the distance from the origin: $\delta(\rho, \phi, \theta) = k\rho$ for constant $k > 0$.



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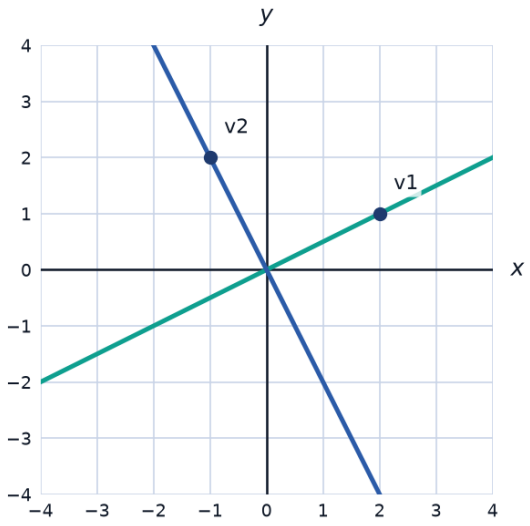
9 **[Intermediate]** • **Target: Distributions & Maximum Likelihood Practice Exercises**

Derive the Gaussian integral $I = \int_{-\infty}^{\infty} e^{-x^2/2} dx = \sqrt{2\pi}$ from first principles by evaluating the double integral $I^2 = \iint_{\mathbb{R}^2} e^{-(x^2+y^2)/2} dA$ in polar coordinates.



10 **[Intermediate]** • **Target: Distributions & Maximum Likelihood Practice Exercises**

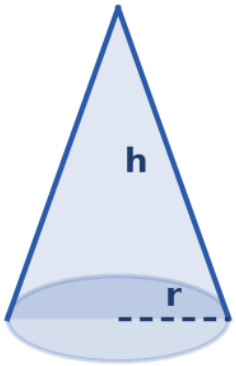
Prove that if $\mathbf{x}_p$ is any particular solution to $A\mathbf{x} = \mathbf{b}$, then the complete solution set is the affine subspace $S = \{\mathbf{x}_p + \mathbf{x}_h : \mathbf{x}_h \in \text{Null}(A)\}$.



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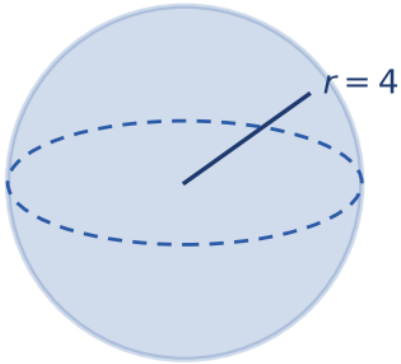
11 [Intermediate] • Target: Distributions & Maximum Likelihood Practice Exercises

Compute the moment of inertia I_z about the z-axis of a solid homogeneous right circular cone of base radius R, height h, and uniform total mass M.



12 [Intermediate] • Target: Distributions & Maximum Likelihood Practice Exercises

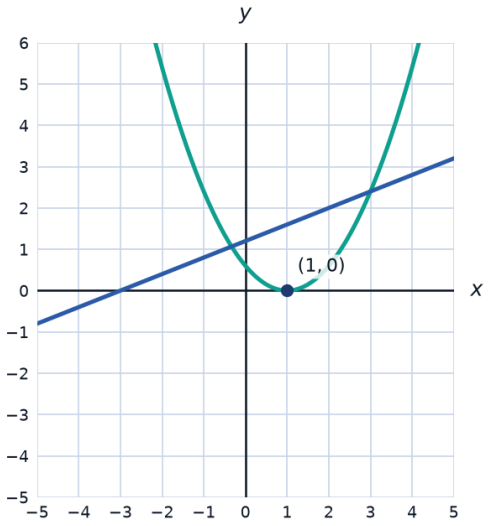
Find the volume and the z-coordinate of the centroid $\bar{z}$ of the solid region bounded below by the xy-plane $z = 0$ and above by the paraboloid $z = 4 - x^2 - y^2$, assuming uniform density $\delta_0 = 1$.



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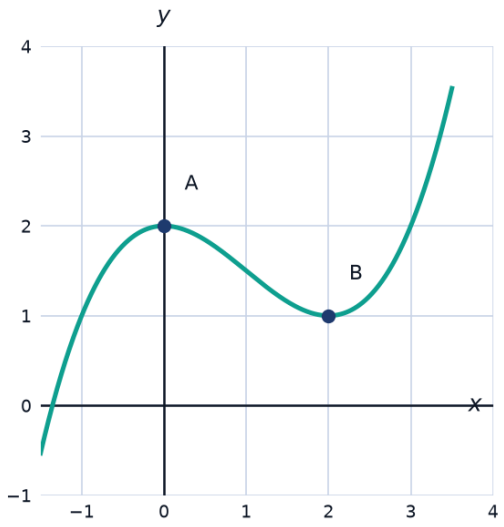
13 [Intermediate] • Target: Distributions & Maximum Likelihood Practice Exercises

Use the Gauss–Jordan elimination algorithm on the augmented block matrix $[A \mid I_3]$ to compute A^{-1} for: $A = (1, 0, 2; 2, -1, 3; 4, 1, 8)$.



14 [Intermediate] • Target: Distributions & Maximum Likelihood Practice Exercises

Evaluate the Cartesian triple integral $\iiint_B xyz^2\,dV$ over the rectangular box $B = [0, 1] \times [-1, 2] \times [0, 3]$.



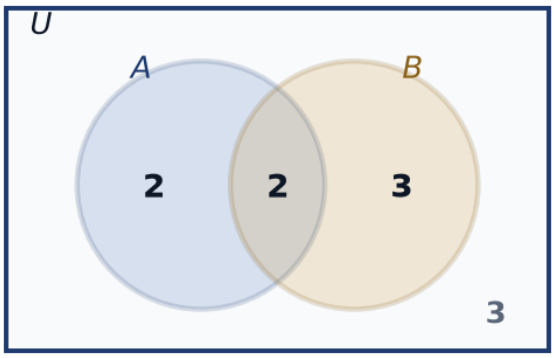
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Distributions & Maximum Likelihood Practice Exercises • Cross-Lesson: 6.01 Linear Regression and OLS

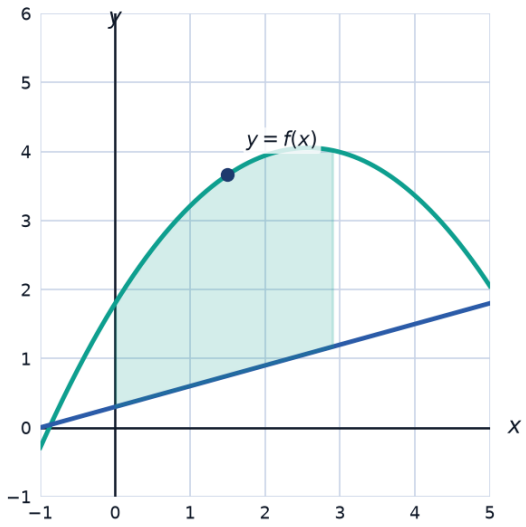
Express the complete solution set of the non-homogeneous linear system in parametric vector form:

$$\{x_1 + 2x_2 - x_3 + 3x_4 = 4;\ 2x_1 + 4x_2 - x_3 + 8x_4 = 11;\ -x_1 - 2x_2 + 2x_3 - x_4 = -1$$



16 [Challenging] • Target: Distributions & Maximum Likelihood Practice Exercises • Cross-Lesson: 7.01 Logistic Regression and Classification

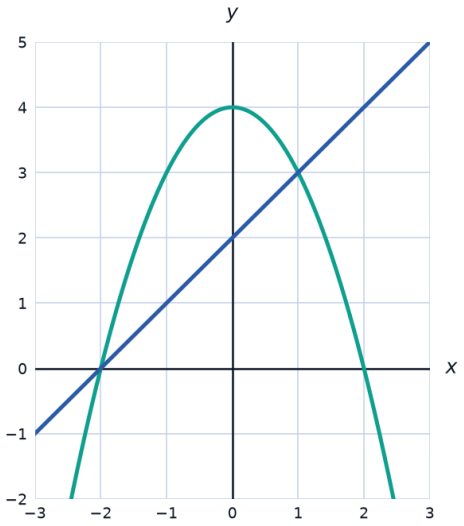
Evaluate the double integral $\iint_R x\cos(x+y)\,dA$ over the rectangular region $R = [0, \pi] \times [0, \pi/2]$.



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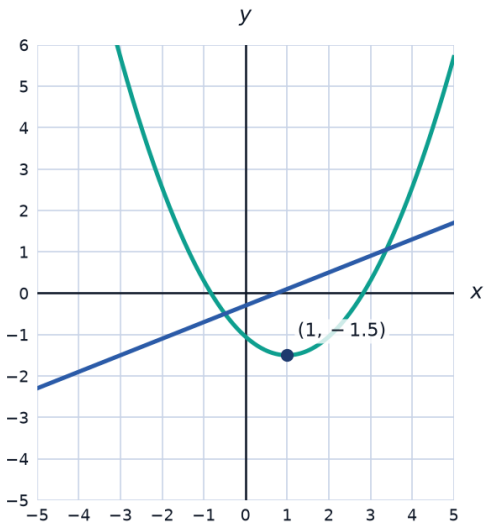
17 **[Challenging]** • **Target: Distributions & Maximum Likelihood Practice Exercises** • **Cross-Lesson: 8.01 PCA and Dimensionality Reduction**

Evaluate the non-elementary integral $\int_0^1 \int_y^1 e^{x^2}\,dx\,dy$ by reversing the order of integration.



18 **[Challenging]** • **Target: Distributions & Maximum Likelihood Practice Exercises** • **Cross-Lesson: 1.01 Vectors and Embeddings in AI**

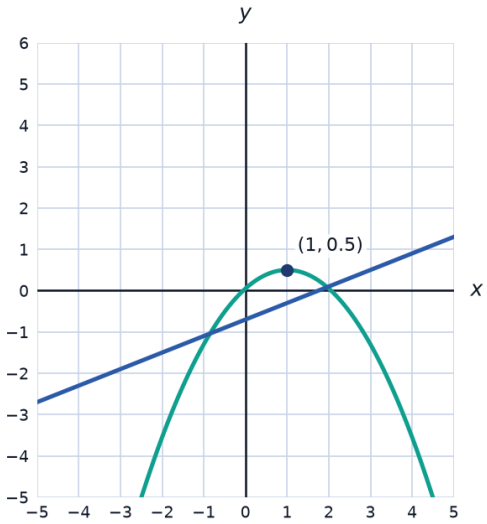
Find the directional derivative of $f(x, y) = x^2y - 3y^2$ at the point $P(1, 2)$ in the direction of the vector $\mathbf{v} = \langle 3, 4 \rangle$.



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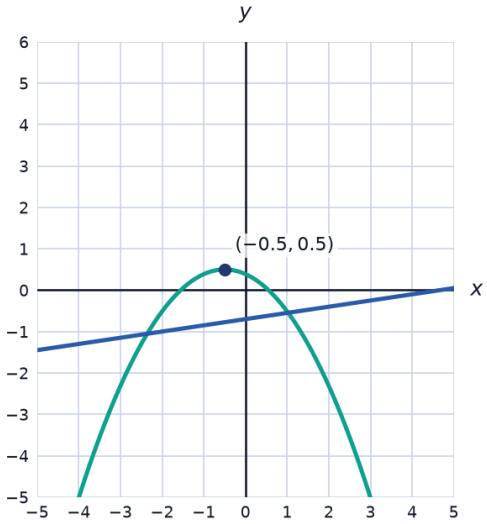
19 [Challenging] • Target: Distributions & Maximum Likelihood Practice Exercises • Cross-Lesson: 2.01 Neural Layer Matrices and Activations

Verify Clairaut's Theorem on equality of mixed partial derivatives for the function $f(x, y) = \cos(2x - 3y) + x^2y^3$ by computing f_{xy} and f_{yx} directly.



20 [Challenging] • Target: Distributions & Maximum Likelihood Practice Exercises • Cross-Lesson: 3.01 Gradient Descent and Optimization

Find the shortest Euclidean distance from the point $P(3, -1, 4)$ to the hyperplane $\mathcal{H} : 2x - y + 2z = 5$ in $\mathbb{R}^3$.



Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Distributions & Maximum Likelihood Practice Exercises

Consider the parametric curve defined by $x(t) = 3\cos(t) - \cos(3t)$ and $y(t) = 3\sin(t) - \sin(3t)$ for $t \in [0, 2\pi]$. This curve traces out a shape known as a 'cardioid'. Find the maximum area of a triangle that can be inscribed within the loop of this cardioid, with one vertex at the origin.

a. $\frac{27\sqrt{3}}{2}$

b. $\frac{81}{2}$

c. $\frac{81\sqrt{3}}{4}$

d. $54\sqrt{3}$

22 [Foundational] • Target: Distributions & Maximum Likelihood Practice Exercises

A marketing research firm is analyzing the relationship between the preferred social media platform and age group of consumers. They collected data from a sample and summarized it in the two-way table below. The marginal distributions are also provided. The research team is interested in understanding how the distribution of preferred platform varies across age groups and is considering a chi-square test for independence. If the null hypothesis of independence is rejected, the observed counts will be significantly different from what would be expected under independence. Assume that a chi-square test for independence is to be performed on this data. Which of the following represents a potential challenge or insight derived from the structure of the data and the test's assumptions?

a. The number of degrees of freedom for the chi-square test can be determined solely from the total number of observations, irrespective of the table dimensions.

b. If the expected count in any cell is less than 5, a normal approximation to the binomial distribution for each cell count would be appropriate to proceed with the chi-square test.

c. The calculation of expected counts under the null hypothesis of independence relies on the marginal proportions of the observed data.

d. A high chi-square statistic indicates that the observed counts are very close to the expected counts, suggesting a strong association between age group and preferred social media platform.

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23 [Foundational] • Target: Distributions & Maximum Likelihood Practice Exercises

A researcher is analyzing the relationship between the number of hours a student spends on extracurricular activities per week and their final exam score in a challenging STEM course. The researcher collects data from 100 students and fits a linear regression model. The model is given by $\text{Exam Score} = 65 + 2.5 \times \text{Hours} - 0.3 \times \text{Hours}^2$, where Exam Score is the final exam score and Hours is the number of hours spent on extracurricular activities per week. The researcher believes that for a certain range of hours, there might be an optimal number of extracurricular hours that maximizes the exam score, and beyond that, the score might decrease due to time constraints. \\ \\ Consider the function representing the exam score as a function of hours: $f(h) = 65 + 2.5h - 0.3h^2$. To find the maximum exam score and the corresponding hours, calculus can be used. \\ Which of the following is the approximate number of extracurricular hours per week that would maximize the student's final exam score, assuming the model holds for $h \geq 0$?

a. 4.17 hours

b. 2.5 hours

c. 5.0 hours

d. 8.33 hours

24 [Foundational] • Target: Distributions & Maximum Likelihood Practice Exercises

A pharmaceutical company is testing a new drug designed to reduce blood pressure. They decide to conduct a randomized controlled experiment. To investigate the potential for confounding variables related to lifestyle, they collect preliminary data on participants' self-reported levels of physical activity and dietary habits. They categorize physical activity into three levels: 'Low', 'Moderate', and 'High'. Dietary habits are categorized into four types: 'Vegetarian', 'Pescatarian', 'Low-Carb', and 'Omnivore'. If the company wants to ensure that any observed association between these two categorical variables in their sample data can be rigorously tested for statistical significance using a chi-square test of independence, and they obtain a contingency table with 3 rows (for physical activity) and 4 columns (for dietary habits), what is the maximum number of independent pieces of information contributing to the chi-square statistic, often referred to as the degrees of freedom, in this specific test?

a. 12

b. 9

c. 6

d. 2

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25 [Foundational] • Target: Distributions & Maximum Likelihood Practice Exercises

What is the maximum rate of change of $f(x, y) = x\cos(y) - 3y$ at $(2,\ 3)$?

a. $3\sin^2(3) + 12\sin(3) + 10$

b. $2\sin(3) - \cos(3) + 3$

c. $\sqrt{3\sin^2(3) + 12\sin(3) + 10}$

d. $2\sin(3) + 3$

26 [Foundational] • Target: Distributions & Maximum Likelihood Practice Exercises

Maximize $f(x, y) = xy$ subject to $x + y = -4$. Find the maximum value.

a. 16

b. 4

c. 5

d. -4

27 [Foundational] • Target: Distributions & Maximum Likelihood Practice Exercises

Let $F(x) = \int_0^x (t^2 + 5t + 4)\,dt$. At which x does F have a local maximum?

a. -1

b. $-\frac{5}{2}$

c. 0

d. -4

28 [Foundational] • Target: Distributions & Maximum Likelihood Practice Exercises

Find the maximum of $f(x, y) = xy$ on the ellipse $\frac{x^2}{4} + \frac{y^2}{25} = 1$.

a. 5

b. 10

c. $\frac{5}{2}$

d. $\frac{7}{2}$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Distributions & Maximum Likelihood Practice Exercises

Find the global maximum value of $f(x) = x^3 - 3x^2$ on $[-1, 3]$.

a. 0

b. 3

c. 2

d. -4

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30 [Intermediate] • Target: Distributions & Maximum Likelihood Practice Exercises

Find the maximum of $f(x, y) = 5x + 12y$ on the circle $x^2 + y^2 = 25$.

a. 85

b. 325

c. 65

d. 169

31 [Intermediate] • Target: Distributions & Maximum Likelihood Practice Exercises

For $f(x) = x^3 - 12x$, find the local maximum value of f.

a. 14

b. 16

c. -16

d. 20

32 [Intermediate] • Target: Distributions & Maximum Likelihood Practice Exercises

A population has $\mu = 141$ and $\sigma = 30$. For a random sample of size $n = 36$, find $P(\bar{x} < 143)$ using the sampling distribution of the sample mean.

a. 0.7554

b. 0.9554

c. 0.3554

d. 0.6554

33 [Intermediate] • Target: Distributions & Maximum Likelihood Practice Exercises

The Central Limit Theorem states that sample means are approximately Normal when:

a. Population is Normal

b. n is large (n≥30)

c. σ is known

d. p<0.5

34 [Intermediate] • Target: Distributions & Maximum Likelihood Practice Exercises

A population proportion is $p = 0.38$. For samples of size $n = 300$, find the standard deviation of the sampling distribution of $\hat{p}$.

a. 0.058

b. 0.028

c. 0.018

d. 0.068

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35 [Intermediate] • Target: Distributions & Maximum Likelihood Practice Exercises

Find the maximum of $x + y + z$ on the sphere $x^2 + y^2 + z^2 = 3$.

a. 3

b. 9

c. $\sqrt{3}$

d. $3\sqrt{3}$

36 [Intermediate] • Target: Distributions & Maximum Likelihood Practice Exercises

A Normal random variable has $\mu = 189$ and $\sigma = 8$. Find $P(173 < X < 213)$.

a. 0.8759

b. 0.9759

c. 0.7759

d. 0.6759

37 [Intermediate] • Target: Distributions & Maximum Likelihood Practice Exercises

Maximum of xy on ellipse $x^2/4 + y^2/9 = 1$ by Lagrange:

a. 3/2

b. 3

c. 9/4

d. 6

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Distributions & Maximum Likelihood Practice Exercises • Cross-Lesson: 1.01 Vectors and Embeddings in AI

Equal squares are cut from the corners of a 24 cm × 24 cm sheet to make an open box. What is the maximum volume (cm^3)?

a. 1024

b. 2048

c. 864

d. 512

39 [Challenging] • Target: Distributions & Maximum Likelihood Practice Exercises • Cross-Lesson: 2.01 Neural Layer Matrices and Activations

For $f(x, y) = x^2 + y^2$ on $x^2 + y^2 \le 1$: global maximum is attained at:

a. boundary $x^2 + y^2 = 1$ (equals 1)

b. origin

c. interior critical point only

d. corner

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40 [Challenging] • Target: Distributions & Maximum Likelihood Practice Exercises • Cross-Lesson: 3.01 Gradient Descent and Optimization

Events A and B are independent with $P(A) = 0.33$ and $P(B) = 0.88$. Find $P(A \cap B)$.

a. 0.4904

b. 0.2904

c. 0.2404

d. 0.1904

41 [Challenging] • Target: Distributions & Maximum Likelihood Practice Exercises • Cross-Lesson: 4.01 Conditional Probability and Bayes Rule

If $f(x) = xe^{-x}$, the maximum value occurs at $x =$

a. -1

b. 0

c. 1

d. e

42 [Challenging] • Target: Distributions & Maximum Likelihood Practice Exercises • Cross-Lesson: 6.01 Linear Regression and OLS

Binomial distribution requires:

a. Continuous outcomes

b. Fixed n, two outcomes, constant p, independence

c. Normal data

d. Dependent trials

43 [Challenging] • Target: Distributions & Maximum Likelihood Practice Exercises • Cross-Lesson: 7.01 Logistic Regression and Classification

A population has $\sigma = 30$. For random samples of size $n = 9$, find the standard deviation of the sampling distribution of $\bar{x}$ (the standard error).

a. 13

b. 6

c. 12

d. 10

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44 [Challenging] • Target: Distributions & Maximum Likelihood Practice Exercises • Cross-Lesson: 8.01 PCA and Dimensionality Reduction

Events A and B satisfy $P(A) = 0.55$, $P(B) = 0.12$, and $P(A \cap B) = 0.0$. Find $P(A \cup B)$.

a. 0.57

b. 0.67

c. 0.82

d. 0.62

45 [Challenging] • Target: Distributions & Maximum Likelihood Practice Exercises • Cross-Lesson: 1.01 Vectors and Embeddings in AI

A binomial random variable has $n = 6$ trials and success probability $p = 0.25$. Find $P(X = 6)$.

a. 0.0044

b. 0.0002

c. 0.1318

d. 0.033

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